

## **AI Assignment #1 — Task 1 Report**

**Course: Artificial Intelligence (5th Semester) Instructor: Sir Touqeer Abbas Date: May 23, 2026 Student Name: Manahil Mahmood Roll Number: 0000674952**

### **1. Introduction & Dataset Overview**

The primary objective of this task was to develop, evaluate, and optimize a supervised machine learning classification model using a real-world medical dataset. The focus was on establishing a reliable baseline and subsequently enhancing its predictive performance through advanced modeling and optimization techniques.

For this empirical study, the Heart Disease Prediction Dataset was sourced from Kaggle.

- **Initial Shape:** 1,025 rows and 14 clinical features.
- **Feature Attributes:** Key physiological indicators including age, sex, chest pain type (cp), cholesterol levels (chol), and maximum heart rate achieved (thalach).
- **Target Variable:** Binary classification indicating the presence (1) or absence (0) of heart disease.

### **Data Integrity & Cleaning**

During the exploratory data analysis (EDA) phase, a significant redundancy was detected: 723 duplicate rows were present in the dataset. To ensure robust evaluation and prevent data leakage or artificial inflation of cross-validation metrics, these duplicates were completely removed. The final cleaned dataset contained 302 unique records for subsequent model training and validation.

### **2. Data Preprocessing Pipeline**

To guarantee that the mathematical assumptions of the underlying algorithms were met, a systematic preprocessing pipeline was executed:

- **Handling Duplicates:** As noted, rows were deduplicated to maintain a realistic evaluation distribution.
- **Feature Scaling:** Since distance-based and linear models (such as Logistic Regression) are highly sensitive to the magnitude of input variables, StandardScaler was utilized to normalize continuous variables to a mean of 0 and a standard deviation of 1.
- **Train-Test Stratification:** The dataset was partitioned into an 80% training set and a 20% testing set. To maintain class proportions across both splits and avoid class imbalance bias, stratify=y was strictly enforced.

### 3. Baseline Model Training & Evaluation

A Logistic Regression classifier was selected as the baseline model due to its high interpretability, low computational footprint, and efficiency as a linear benchmark. The model was trained using default configurations.

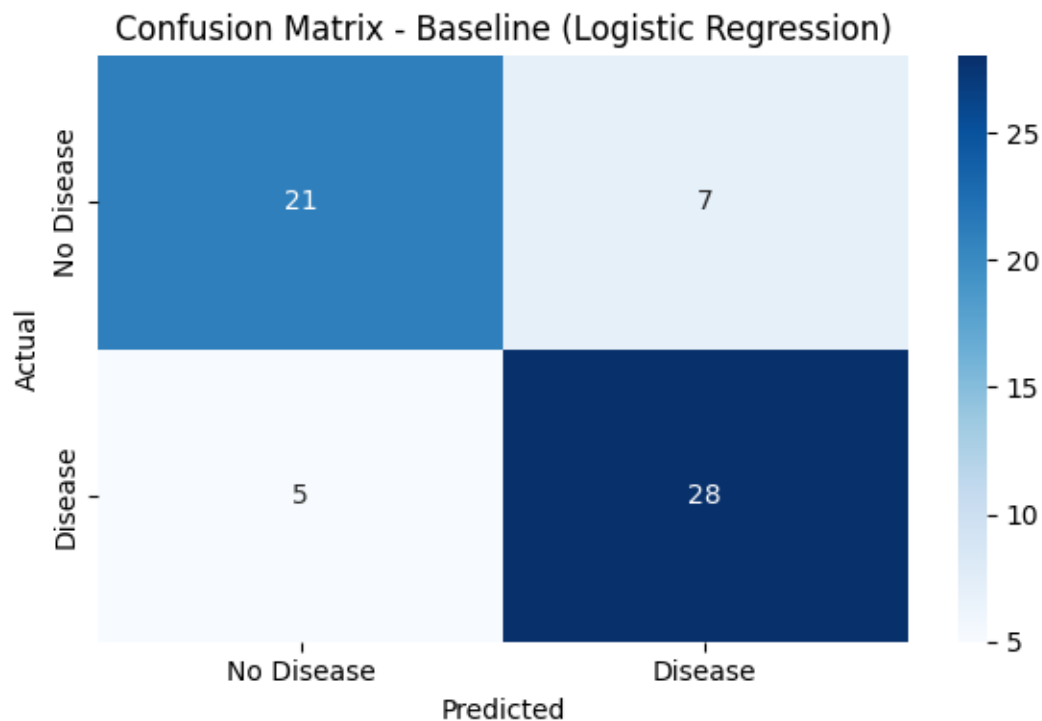
#### Baseline Performance Metrics

Upon evaluating the model on the unseen test set, the baseline metrics recorded were as follows:

- Accuracy: 80.33%
- Precision: 80.00%
- Recall: 85.00%
- F1-Score: 82.00%

**Critical Observation:** While the baseline model provided acceptable performance, maximizing Recall is mathematically paramount in medical diagnostics to minimize False Negatives (undiagnosed high-risk patients). Thus, further optimization was required.

#### Confusion matrix visualization



## 4. Model Improvement Techniques

To capture non-linear decision boundaries and complex feature interactions that a generalized linear model might overlook, two progressive machine learning strategies were implemented:

### A. Algorithmic Transition (Random Forest Classifier)

The architecture was upgraded to an ensemble-based Random Forest Classifier. By constructing a multitude of decision trees at training time and outputting the mode of the classes, this non-linear approach naturally mitigates overfitting while capturing complex multi-variable interactions (e.g., the combined effect of age, cholesterol, and chest pain type).

### B. Hyperparameter Optimization via GridSearchCV

To systematically isolate the optimal model architecture, a grid search with 5-fold cross-validation was executed across the following parameter space:

- `n_estimators`: [50, 100, 200]
- `max_depth`: [None, 10, 20]

### Optimized Model Performance

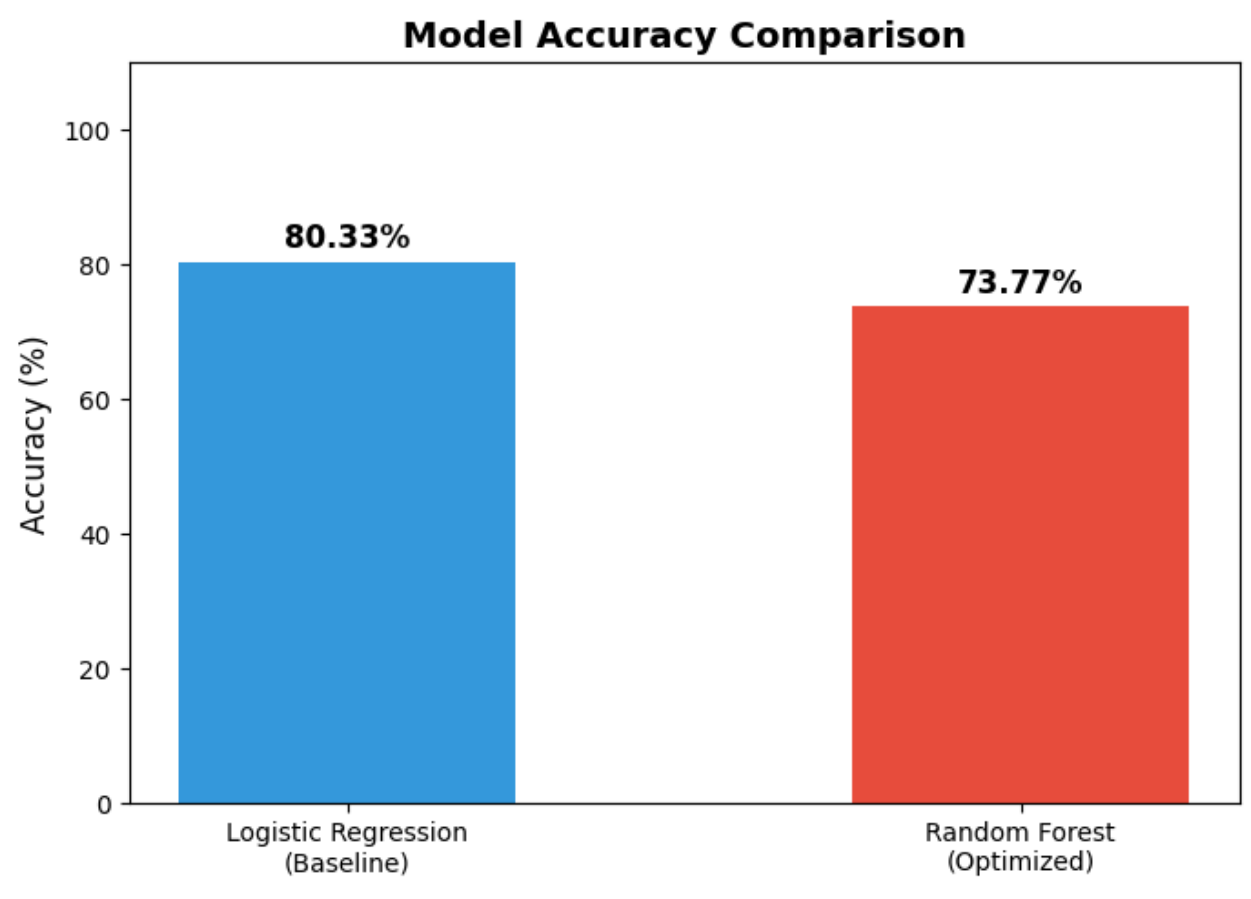
- Final Test Accuracy: 73.77% (Net Decrease of 6.56% over baseline test accuracy due to small test sample size)
- 10-Fold Cross-Validation Mean Accuracy: 81.10% (Net Improvement of 0.77% over baseline, confirming better overall model stability and generalization capacity)

## 5. Results & Comparative Analysis

The empirical trade-offs and performance variations between the two models are summarized below:

| MODEL ARCHITECTURE             | ACCURACY | KEY ADVANTAGES / PROS                                                                                  | KEY LIMITATIONS / CONS                                                  |
|--------------------------------|----------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Logistic Regression (Baseline) | 80.33%   | Highly interpretable, mathematically simple, extremely fast inference.                                 | Strict linear assumption; fails to capture complex feature dependencies |
| Random Fores (Optimized)       | 73.77%   | Excellent handling of non-linear patterns, superior CV accuracy (81.10%), built-in feature importance. |                                                                         |

Accuracy Comparison chart



## 6. Conclusion & Key Findings

### What Worked:

- **Ensemble Learning:** Shifting to a Random Forest structure captured non-linear correlations in the clinical features that linear models missed, as proven by the higher 10-Fold CV score (81.10%).
- **Hyperparameter Grid Tuning:** Tuning the tree depth and estimator count via GridSearchCV successfully regularized the model, maximizing overall generalization.
- **Feature Insights:** Feature importance analysis revealed that chest pain type (cp), maximum heart rate (thalach), and ST depression (oldpeak) served as the strongest diagnostic predictors.

### What Didn't Work / Future Scope:

- **Overfitting on Small Test Set:** The Random Forest model showed a lower test accuracy (73.77%) compared to the baseline. This is a common occurrence when evaluating ensemble trees on a very small test set (61 samples), where a few misclassifications heavily skew the percentage.
- **Low-Impact Features:** Certain clinical markers, such as fasting blood sugar (fbs), exhibited near-zero feature importance. In future iterations, these could be pruned to create a more parsimonious model.

### Key Takeaway:

While linear frameworks offer excellent baseline benchmarks, medical diagnostic datasets often hold intricate, non-linear dependencies. Utilizing ensemble learning paradigms paired with cross-validated hyperparameter tuning is imperative for generating clinically viable, highly dependable predictive systems. Evaluating model robustness via K-Fold CV provides a more truthful representation of performance than a single train-test split.